

3a	Centripetal acceleration: <u>Car on path X and Y has the same acceleration</u> Both experienced the <u>same centripetal force at maximum speed</u> since the <u>centripetal force is provided by the (maximum lateral) friction force</u>. As both cars are identical and hence have the <u>same mass</u>, they will have the <u>same centripetal acceleration</u>. Maximum speed: <u>Car on path Y will have a bigger maximum speed.</u> $F = \frac{mv^2}{r}$ $v = \sqrt{\frac{Fr}{m}}$ Since F and m are the same for both cars, $v \propto \sqrt{r}$. Since radius of path Y is bigger than that of X, maximum speed of car on path Y is larger.	A	M1 A1 M1 A1
3bi	For circular motion, centripetal force on moon of mass m is provided by the gravitational force due to Jupiter on moon, $\frac{GM_J m}{R^2} = mR\omega^2$ $G \frac{M_J m}{R^2} = mR\left(\frac{2\pi}{T}\right)^2$ $\Rightarrow T = \sqrt{\frac{4\pi^2 R^3}{GM_J}}$ where ω is the angular velocity of moon.	A	M1 M1 A0
3bii	Since $T^2 \propto R^3$, $\left(\frac{T_{Th}}{T_{Am}}\right)^2 = \left(\frac{R_{Th}}{R_{Am}}\right)^3 \Rightarrow \left(\frac{0.676}{T_{Am}}\right)^2 = \left(\frac{3.18}{2.62}\right)^3$ $T_{Am} = 0.506$ Earth-days	A	M1 A1

3biii	Since the <u>orbital period</u> of Thebe is not the same as the <u>rotation period</u> of Jupiter, So not in stationary orbit. Or orbital period of Thebe = 0.676 Earth-days, not the same as a <u>Jupiter-day which is approximately 0.417 Earth-days</u>, so not (possible to be) in stationary orbit.	D	M1 A1 (M1) (A1)
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